Guilherme José

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Education

- **University of Amsterdam / Vrije Universiteit (UvA/VU)** — Amsterdam, Netherlands
  MSc in Computer Science (Joint Degree) – Track: Computer Systems and Infrastructure — *September 2026 – July 2028*
- **Instituto Superior Técnico** — Lisbon, Portugal
  Bachelor in Computer Science and Engineering; Grade: 16/20 — *September 2023 – July 2026*
  - **Coursework**: Linear Algebra, Calculus 1, 2 and 3, Probability and Statistics, Discrete Mathematics, Physics 1 and 2, Algorithms and Data Structures, Object-Oriented Programming, Operating Systems, Databases, Machine Learning, Artificial Intelligence, Computer Networks, Computer Architecture, Computer Organization, Distributed Systems, Software Engineering, Compilers, Computer Graphics
- **Grenoble INP - ENSIMAG** — Grenoble, France
  MIAI Summer School in Artificial Intelligence — *July 2024*

Experience

- **SingleStore** — Lisbon, Portugal
  Software Engineering Intern - Engine Department — *June 2026 – August 2026*
  - **Clustering**: Developed and maintained clustering features within the distributed database engine, supporting node coordination and system stability.
  - **Two-Phase Commit (2PC)**: Implemented a feature extension for the 2PC protocol that triggers during failure scenarios, ensuring fault tolerance and consistent state recovery across distributed nodes.
  - **Concurrency Monitoring**: Engineered monitoring capabilities and applied optimizations to internal engine spin locks, improving visibility into thread contention and concurrency metrics.
  - **Technologies Used**: C, C++, Python
- **BNP Paribas** — Lisbon, Portugal
  Summer Internship - Quantitative Research - Compute Team — *July 2025 – August 2025*
  - **Website**: Built a full-stack website serving as a wrapper for an external API, improving usability and accessibility of the service.
  - **Automation Tasks**: Developed automation scripts to streamline repetitive tasks, reducing manual effort and increasing efficiency.
  - **Technologies Used**: Python, Flask

Projects

- **K-means implementation RISC-V**: Implemented a 2D K-Means clustering algorithm from scratch, performing cluster assignment and iterative centroid updates using assembly for the RISC-V architecture. **View repository.**
- **Go Game**: Developed a terminal-based implementation of the board game Go in Python, featuring a complete ruleset and terminal based interface. **View repository.**
- **SQL Problem Solving**: Solved the Harvard CS50 Week 7 problem "Fiftyville," analyzing a SQLite database of fictional city data to identify a thief by examining bank accounts, flights, and passport records. **View repository.**
- **Github Workflow Monitor**: Implemented a command line tool in Java that fetches workflow runs, jobs and steps via GitHub API and prints them to stdout. **View repository.**

Awards & Competitions

- **SWERC 2025**: Participant in Southwestern Europe Regional Contest (University-level competitive programming).
- **MIUP 2025**: Participant in Portuguese national university competitive programming contest.

Programming Skills

- **Languages**: C/C++, Rust, Java, Python, SQL
- **Technologies**: Git, Docker, gRPC, Spring Boot, Vue.js, scikit-learn, Pandas, ROS
- **Concepts/Methods**: Data Structures, Algorithms, Distributed Systems, OOP, Linux

Languages

- **Portuguese:**: Mother tongue
- **English:**: Cambridge C1 (Advanced)